

B.Tech III Year II Semester (R20) Supplementary Examinations November/December 2024

CLOUD COMPUTING

(Common to CSE (IOT) & CSE (CS))

Time: 3 hours

Max. Marks: 70

PART – A

(Compulsory Question)

- 1 Answer the following: (10 X 02 = 20 Marks)
- | | |
|---|----|
| (a) Enlist the applications of Cloud Computing. | 2M |
| (b) Define Access Management for Cloud System. | 2M |
| (c) Define Approaches to execute Hadoop Map Reduce Job Execution. | 2M |
| (d) What are the different types of Data Structure? | 2M |
| (e) Which is the Google Cloud Platform? | 2M |
| (f) What are the applications of Windows Azure? | 2M |
| (g) Which type of data is called as Big Data? | 2M |
| (h) Differentiate between Structured, Unstructured and Quasi Structured Data. | 2M |
| (i) Is Virtualization a good practice to implement? Justify your answer. | 2M |
| (j) List the change Management Models in Cloud Computing. | 2M |

PART – B

(Answer all the questions: 05 X 10 = 50 Marks)

- | | | |
|-----------|---|-----|
| 2 | Explain Cloud Computing Cube Model, Service models and Deployment Models. | 10M |
| OR | | |
| 3 | How to reduce the operation overheads in Cloud Computing? | 10M |
| 4 | Explain Cloud Application Design Methodologies in Python. | 10M |
| OR | | |
| 5 | Explain the Hadoop Architecture used for Cloud Computing. | 10M |
| 6 | Explain the design Approach for building Cloud Framework. | 10M |
| OR | | |
| 7 | Explain the working of RESTFUL web API. | 10M |
| 8 | What are the applications for Multimedia Cloud? Explain any two of them. | 10M |
| OR | | |
| 9 | Define Fed RAMP, Cloud Audit, ENISA Cloud. | 10M |
| 10 | Describe the Data Security Tools and Techniques. | 10M |
| OR | | |
| 11 | Describe Cloud Forensics and Key Management in Cloud. | 10M |

B.Tech III Year II Semester (R20) Supplementary Examinations November/December 2024

CLOUD COMPUTING

(Common to IT, AIDS, CSE (AI), CSE (AI&ML) and CSE (DS))

Time: 3 hours

Max. Marks: 70

PART – A
(Compulsory Question)

1 Answer the following: (10 X 02 = 20 Marks)

- | | |
|--|----|
| (a) List the Cloud Based Services. | 2M |
| (b) Define Replication. | 2M |
| (c) What are the Data Storage Approaches? | 2M |
| (d) List the various packages for Python required to implement Cloud Applications. | 2M |
| (e) List the steps in designing REST full Web API. | 2M |
| (f) How can Python be used to access Amazon Web Services. | 2M |
| (g) Define Big Data. | 2M |
| (h) Draw the Big Data Analytics Life Cycle. | 2M |
| (i) List the Applications of Cloud Computing. | 2M |
| (j) List two approaches for migrating data to cloud. | 2M |

PART – B

(Answer all the questions: 05 X 10 = 50 Marks)

- 2 What are the key features of information governance? What are the policies and procedures used in data security to enforce information governance in cloud environment. 10M

OR

- 3 How can the performance of complex processing of large datasets be increased through scaling out in Cloud? 10M

- 4 Describe the architecture of map Reduce Framework. How is the Hadoop different from the other data processing tools? 10M

OR

- 5 Explain the Data Storage Approaches used in Cloud Computing. 10M

- 6 What role does python play in developing Google Cloud Platform? 10M

OR

- 7 Describe the Map Reduce Process. 10M

- 8 Compare the EMC cloud offering with the cloud service provided by Amazon, Google and Microsoft. 10M

OR

- 9 Explain Load Testing and Bottleneck Detection in Hadoop Benchmarking. 10M

- 10 Explain the issues raised by implementation of Virtualization in Cloud Computing. 10M

OR

- 11 What are the legal Issues in Cloud Computing? Explain in brief. 10M

B.Tech IV Year I Semester (R20) Regular & Supplementary Examinations December/January 2024/2025

CLOUD COMPUTING
(Common to CSE, CS&D and AI&ML)

Time: 3 hours

Max. Marks: 70

PART – A
(Compulsory Question)

- 1 Answer the following: (10 X 02 = 20 Marks)
- | | |
|--|----|
| (a) List all the service models available in cloud computing. | 2M |
| (b) What is the role of virtualization in cloud computing? | 2M |
| (c) Discuss about Apache Hadoop, and what are its core components? | 2M |
| (d) How do dictionaries work in Python, and when would you use? | 2M |
| (e) What libraries can be used in Python for Google Cloud Platform (GCP)? | 2M |
| (f) Write any two common design approaches for cloud applications in Python. | 2M |
| (g) Define Big Data Analytics. | 2M |
| (h) Write any two key features of a live video streaming application. | 2M |
| (i) What is the purpose of auditing in cloud security? | 2M |
| (j) Discuss the legal issues in Cloud Computing. | 2M |

PART – B
(Answer all the questions: 05 X 10 = 50 Marks)

- 2 (a) Define scalability and elasticity in the context of cloud computing. How are they different? 5M
- (b) What are Service Level Agreements (SLAs) in cloud computing, and why are they important? 5M

OR

- 3 (a) Describe cloud-based application services, and how do they enhance software development and deployment. 5M
- (b) What are Identity and Access Management (IAM) services in cloud computing, and why are they critical for security? 5M
- 4 (a) What are Hadoop schedulers, and why are they important for resource management in a Hadoop cluster? 5M
- (b) Discuss different data storage approaches in cloud computing and how are they used. 5M

OR

- 5 (a) Explain the role of the Job Tracker and Task Tracker in Hadoop's MapReduce execution process. 5M
- (b) What is a Python package, and how does it differ from a module? Provide an example of how to create a package. 5M
- 6 (a) Describe how Python can be used to interact with Azure services like Virtual Machines and Azure Functions. 5M
- (b) What is the role of Python in implementing MapReduce tasks for Big Data processing? Provide an example. 5M

OR**Contd. in Page 2**

- 7 (a) What are the key principles of designing a RESTful web API in Python, and which Python frameworks support it? 5M
(b) What are the main components of a social media analytics application, and how does it gather and process data? 5M
- 8 (a) Define digital data? Explain in brief about classification of digital data. 5M
(b) What are streaming protocols? Discuss various types of streaming protocols. 5M
- OR**
- 9 (a) What are Application Performance Metrics? Explain the significance of at least three key application performance metrics. 5M
(b) What is Benchmarking? Identify and describe the different types of benchmarking tools used in performance evaluation. 5M
- 10 (a) Write Short notes on: 5M
(i) Identity Access Management,
(ii) Data Security.
(b) Discuss seven step models for cloud migration in detail with neat diagram. 5M
- OR**
- 11 (a) Discuss the impact of cloud computing on the education sector and how it enhances learning and collaboration. 5M
(b) Explain how data location and virtualization in cloud computing raise jurisdictional issues, and why this is a legal concern. 5M
